BITWISE OPERATORS

Compliment(~)

In compliment operator user need to add +1 if the value is in +positive , Need to substract -1 if the value is in -negitive.

In [4]:
```
a = 25
```

In [5]:
```
~a
```

Out[5]:
```
-26
```

In [6]:
```
~35
```

Out[6]:
```
-36
```

In [7]:
```
~-33
```

Out[7]:
```
32
```

AND OPERATOR (&)

In 'AND' operator we dont calculate numbers system

In [17]:
```
25&32
```

Out[17]: 0

In [12]:
```
bin(25)
```

Out[12]: '0b11001'

In [13]:
```
bin(32)
```

Out[13]: '0b100000'

In [16]:
```
int(0b11001)
```

Out[16]: 25

OR operator(|)

In [18]:
```
bin(25)
```

Out[18]: '0b11001'

In [19]:
```
bin(32)
```

Out[19]: '0b100000'

In [20]:
```
12&20
```

Out[20]: 4

In [25]:
```
28&41
```

Out[25]: 8

In [26]:
```
int(28&41)
```

Out[26]: 8

X or XCLUSIVE or (^)

0^0 = 0 1^1 = 1 0^1 = 1 1^0 = 1 1^1 = 0

In [29]:
```
12^13
```
Out[29]: 1

In [30]:
```
25^30
```
Out[30]: 7

In [31]:
```
25^32
```
Out[31]: 57

** LEFT SHIFT (<<) **

In left shift we gain the bit

In [32]:
```
12<<1
```
Out[32]: 24

In [33]:
```
12<<3
```
Out[33]: 96

In [34]:
```
21<<2
```
Out[34]: 84

In [35]:
```
15<<3
```

Out[35]: 120

RIGHT SHIFT

In right shift we loose the bit

In [36]:
```
12>>1
```

Out[36]: 6

In [37]:
```
12>>2
```

Out[37]: 3

In [40]:
```
21>>4
```

Out[40]: 1

NUMBER SYSTEMS

BINARY NUMBER SYSTEM

converts numbers to binary number system

In [41]:
```
bin(25)
```

Out[41]: '0b11001'

In [42]:
```
int(0b11001)
```

Out[42]: 25

In [43]:
```
bin(45)
```

Out[43]: '0b101101'

In [48]:
```
type(0b1101)
```

Out[48]: int

In [47]:
```
int(0b101101)
```

Out[47]: 45

OCTAL NUMBER SYSTEM

In [49]:
```
oct(25)
```

Out[49]: '0o31'

In [50]:
```
int(0o31)
```

Out[50]: 25

In [51]:
```
oct(32)
```

Out[51]: '0o40'

In [52]:
```
oct(41)
```

Out[52]: '0o51'

DECIMAL NUMBER SYSTEM

In [56]:
```
a = 23.4
b = 34.2
```

In [57]:
```
a+b
```

Out[57]:
```
57.6
```

In [59]:
```
a1 = 0xA
a2 = 0xf
print(a1+a2)
```

```
25
```

In [60]:
```
x1 = 0x1A
x2 = 0x5
print(x1+x2)
```

```
31
```

In [63]:
```
int(0x2A)
```

Out[63]:
```
42
```

In [66]:
```
hex(0x21)
```

Out[66]:
```
'0x21'
```

PRINT() FORMAT METHOD

In [67]:
```
x1=20
x2=30
add=x1+x2
```

In [70]:
```
print(x1+x2)
```

```
50
```

In [71]:
```
print(x1-x2)
```

```
-10
```

In [72]:
```
print(x1/x2)
```

```
0.6666666666666666
```

PRINT() SHORT FORMAT METHOD

In [73]:
```
x1=20
x2=30
add= x1+x2
```

In [79]:
```
print((x1+x2))
```

```
50
```

In [ ]: